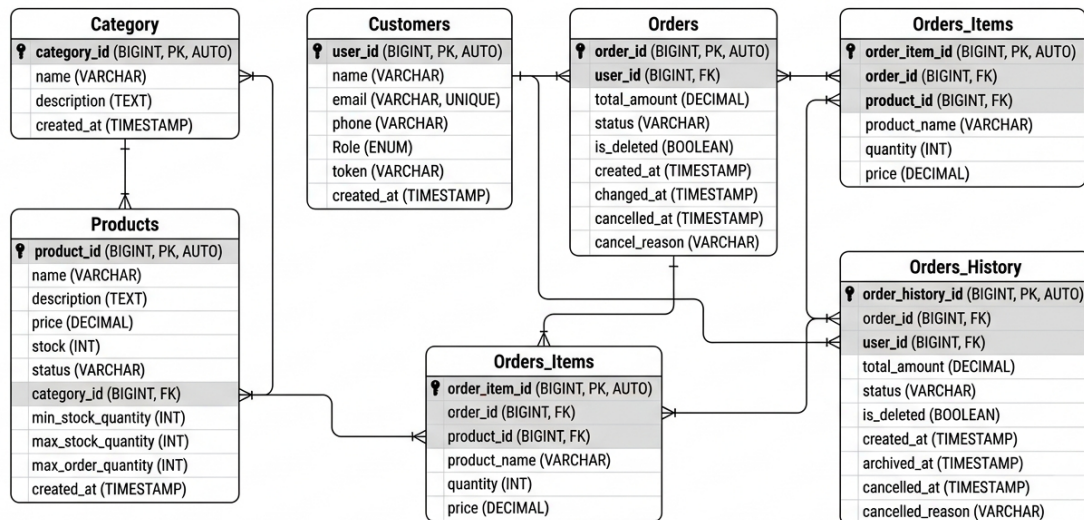


CloudMart Database Schema

Concise and presentable explanation of the provided ER diagram.

ECOMMERCE DATABASE SCHEMA



1. Schema Overview

The database is centered on **Customers, Products, and Orders**. Categories organize products, Orders_Items connects orders with products, and Orders_History keeps historical order information.

2. Tables and Purpose

Table	Purpose	Important Fields
Category	Stores product categories.	category_id (PK), name, description
Customers	Stores customer identity and role information.	user_id (PK), name, email, phone, Role, token
Products	Stores product details, price, stock and quantity limits.	product_id (PK), category_id (FK), price, stock, min/max stock, max order
Orders	Stores the main order record and current status.	order_id (PK), user_id (FK), total_amount, status, timestamps
Orders_Items	Stores products and quantities within each order.	order_item_id (PK), order_id (FK), product_id (FK), quantity, price
Orders_History	Stores historical snapshots of orders.	order_history_id (PK), order_id (FK), user_id (FK), status, timestamps

3. Primary and Foreign Keys

Primary Key (PK): uniquely identifies a row, e.g. Customers.user id and Orders.order id.

Foreign Key (FK): links a row to another table, e.g. Products.category_id → Category.category_id and Orders.user_id → Customers.user id.

4. Key Relationships

Relationship	Meaning
Category → Products	One category can contain multiple products.
Customers → Orders	One customer can place multiple orders.
Orders → Orders_Items	One order can contain multiple items.

Relationship	Meaning
Products → Orders_Items	One product can appear in multiple order items.
Orders → Orders_History	An order can have multiple history records.
Customers → Orders_History	A customer can be referenced by multiple history records.

5. Order Data Flow

Customer → places an **Order** → contains **Orders_Items** → references **Products** → each product belongs to a **Category**.

Orders → **Orders_History** records historical order information and changes.

6. Important Design Points

- Orders_Items connects Orders and Products while storing quantity and price.
- Products maintains stock and ordering limits.
- Orders stores the current order status and cancellation details.
- Orders_History preserves historical order information.
- Customers stores the user role used to distinguish customer/admin access.

7. Schema at a Glance

Category → Products → Orders_Items ← Orders ← Customers
 Orders → Orders_History

CloudMart Database Schema • Based on the provided ER diagram